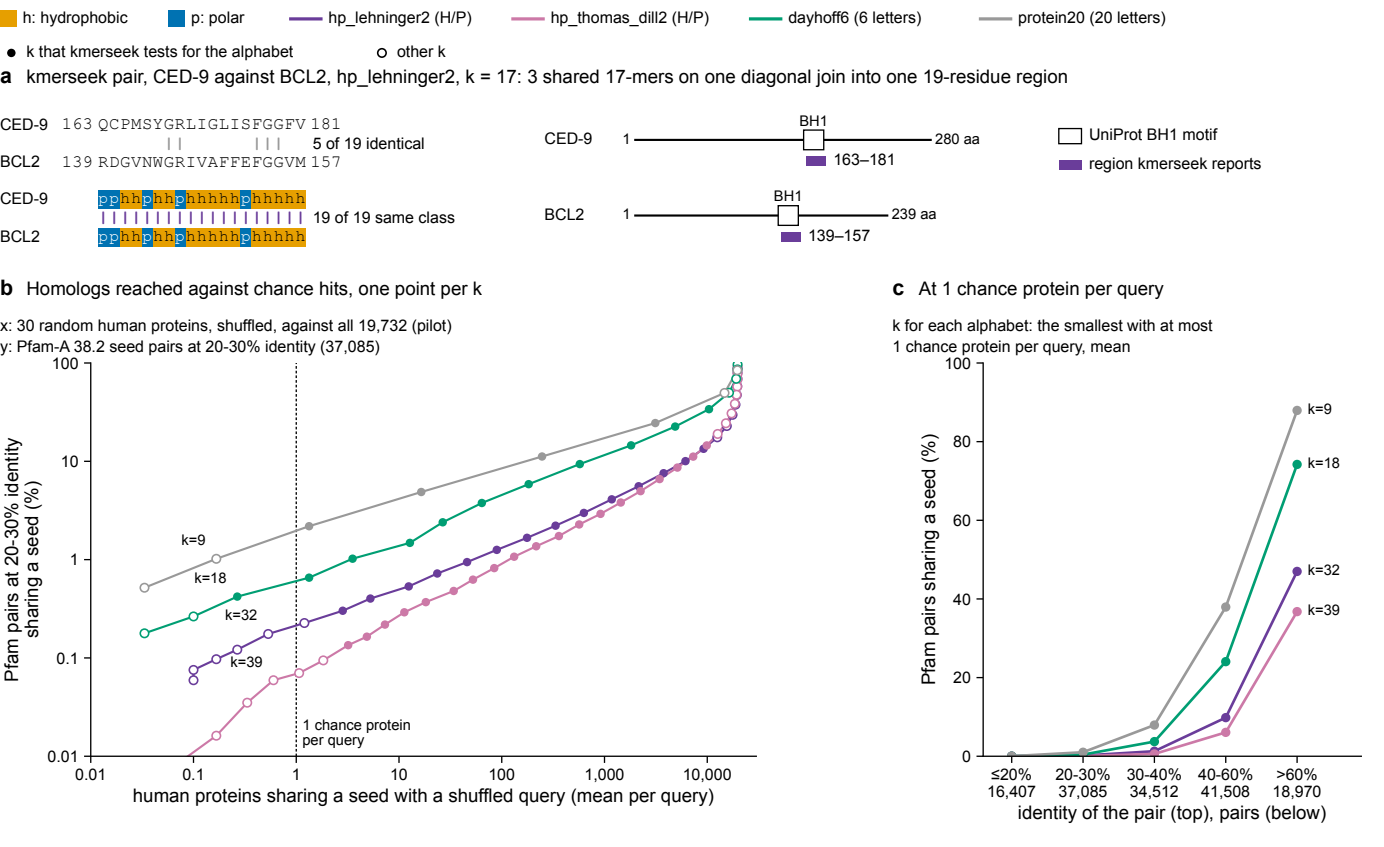

h: hydrophobic
p: polar
hp_lehninger2 (H/P)
hp_thomas_dill2 (H/P)
dayhoff6 (6 letters)
protein20 (20 letters)
k that kmerseek tests for the alphabet
other k
a kmerseek pair, CED-9 against BCL2, hp_lehninger2, k = 17: 3 shared 17-mers on one diagonal join into one 19-residue region
CED-9 163 QCPMSYGRLIGLISFGGFV 181
5 of 19 identical
BCL2 139 RDGVNWGRIVAFFEFGGVM 157
CED-9 ppphhpphhghhhhhghhhhh
19 of 19 same class
BCL2 ppphhpphhghhhhhghhhhh
BH1
CED-9 1 280 aa
163–181
BH1
BCL2 1 239 aa
139–157
UniProt BH1 motif
region kmerseek reports
b Homologs reached against chance hits, one point per k
x: 30 random human proteins, shuffled, against all 19,732 (pilot)
y: Pfam-A 38.2 seed pairs at 20-30% identity (37,085)
Pfam pairs at 20-30% identity sharing a seed (%)
100
10
1
0.1
0.01
k=9
k=18
k=32
k=39
1 chance protein per query
0.01
0.1
1
10
100
1,000
10,000
human proteins sharing a seed with a shuffled query (mean per query)
c At 1 chance protein per query
k for each alphabet: the smallest with at most 1 chance protein per query, mean
Pfam pairs sharing a seed (%)
100
80
60
40
20
0
k=9
k=18
k=32
k=39
≤20% 16,407
20-30% 37,085
30-40% 34,512
40-60% 41,508
>60% 18,970
identity of the pair (top), pairs (below)